

When a high-mountain slope failure cascades downstream: reconstructing the 26 August 2026 Gyirong mixed rock–ice disaster

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Abstract. On 26 August 2026, a high-elevation slope failure involving rock and glacier ice near the China–Nepal border developed into a channelised mass-flow and flood-affected cascade that reached Gyirong Port/Timure. We reconstructed this source–pathway–receptor system from open and publicly released Earth-observation records, combining event-bracketing Sentinel-2, Landsat-9, PlanetScope and Sentinel-1 observations, Copernicus DEM terrain analysis, and UNOSAT and Copernicus Emergency Management Service rapid mapping. Nested multisensor envelopes constrained the post-event changed/source surface to a preferred area of 1.009 km² (conservative–inclusive range 0.491–1.841 km²). A terrain- and drainage-constrained representative route extended 21.843 km in plan to Gyirong Port and descended approximately 3396 m; it is a corridor descriptor, not a particle trajectory. The prespecified pre-event Sentinel-2 sequence showed no unambiguous large-scale surface change, and the final usable optical observation preceded failure by 45.85 h. Neither GPM IMERG V07 Late nor ERA5-Land placed any of five prespecified precipitation windows in an unusually wet upper tail, although their amounts and historical ranks differed strongly. The 7 d ERA5-Land mean 2 m temperature (9.43 °C) and positive-degree-day total (65.94 °C d) ranked above all 25 matched years from 2001–2025. These gridded values provide regional thermal context and do not identify a trigger. Longer-term records support anthropogenic influence on regional summer climate and growth of downstream built-up exposure, while any mechanical link to the rupture remains unresolved. Independent rapid mapping delineated a 37.353 km² downstream disturbed footprint and contained 695 building points and 15.671 km of roads graded destroyed; these are preliminary mapped features rather than a complete damage census. The principal contribution is an uncertainty-explicit, reproducible reconstruction of the full cascade and its observation-scale limits: open Earth observation recovered robust first-order source, propagation, environmental-context and impact constraints, while the exact detached perimeter, source volume, short-lived pre-failure activity, initiation mechanics and source-scale hydrology remained unresolved.

1 Introduction

High-mountain rock–ice cascades are source–pathway–receptor systems. A failure involving bedrock, glacier ice, seasonal snow or unconsolidated sediment may fragment, entrain valley-floor material, enter a confined drainage, impound or displace water, and transform into a debris flow or sediment-laden flood. The resulting downstream footprint can greatly exceed the initial source area (Schneider et al., 2011; Mani et al., 2023; Zhong et al., 2025). Understanding such an event therefore requires more than mapping a scar: the source, propagation corridor, process transitions and affected receptors must be reconstructed together.

Complete reconstruction is difficult because the components are observed by different systems and at different times. Source areas are remote and steep; cloud, seasonal snow and terrain shadow obscure optical imagery; radar responses are affected by viewing geometry and surface state; and rapid-response maps record selected consequences rather than a complete process history. Field access may also be delayed, and a single cascade can cross administrative or national boundaries. These conditions make uncertainty representation and separation of direct observation from interpretation central scientific requirements rather than secondary reporting details.

Recent events demonstrate the diversity of source conditions and cascade evolution. The 2016 Aru events were low-angle glacier detachments associated with surge-like instability, whereas repeated Flat Creek detachments involved a dynamically constrained glacier with a soft bed and high meltwater availability (TIAN et al., 2017; Kääb et al., 2018; Gilbert et al., 2018; Jacquemart et al., 2020). At Chamoli in 2021, steep bedrock and glacier ice fragmented and entrained material before evolving into a destructive debris flow; later work also documented long-term motion and cracking (Shugar et al., 2021; de Vries et al., 2022). Repeated Sedongpu events show how glacier and rock–ice failures can produce exceptional erosion, river blockage and landscape reorganisation (An et al., 2022; Zhao et al., 2022; Kääb and Girod, 2023; Li et al., 2024; Zhang et al., 2025). Together, these cases show that source type, entrainment, process transformation and downstream effects vary substantially, and that glacier collapse, rock avalanche, glacier detachment and glacial-lake outburst flood are not interchangeable labels.

Environmental context, source preconditioning and immediate triggering also require different evidence. Permafrost degradation, glacier retreat, extreme warmth and liquid water are physically relevant to high-mountain slope stability, but their roles depend on local lithology, fracture structure, thermal state and hydrology, which are seldom observed directly (Gruber et al., 2004; Huggel, 2009; Huggel et al., 2010b, a; Gruber et al., 2017). An environmental anomaly can be consistent with preconditioning without identifying the immediate trigger. Likewise, regional anthropogenic climate attribution does not establish event attribution, and historical development can increase exposure without amplifying hazard magnitude. These distinctions are essential when regional warming, unresolved source mechanics and changing settlement patterns coexist (Kargel et al., 2016; Fujita et al., 2017; Zhuang et al., 2024; Jiang and Zhou, 2023; Pesaresi et al., 2024).

Multi-source Earth observation can nevertheless recover complementary parts of an inaccessible cascade. Optical imagery can locate changed surfaces and map snow, ice, debris and channel disturbance; radar amplitude can provide an independent all-weather change signal; digital elevation models can constrain source and route geometry; reanalysis and satellite precipitation can describe regional environmental conditions; and rapid maps can document selected downstream effects. Pre-failure surface motion or morphological change is one additional observational dimension, but its interpretation depends on sensor resolution, cadence, cloud, snow, texture and terrain geometry. The value of a multisensor reconstruction is therefore greatest when direct observations, derived quantities, supported interpretations and unresolved mechanisms remain explicit (Huggel et al., 2005; Shugar et al., 2021; Zhao et al., 2022; Kirschbaum et al., 2019).

For Gyirong, Guo et al. (2026) provide the event-specific process benchmark. They reconstructed the source-to-port chain and progressive entrainment using GF-1/GF-7 and other satellite imagery, field photographs, KKN seismic records, monitoring video, eyewitness constraints and a dedicated 2024–2026 Sentinel-2 glacier-velocity analysis. Their study reported limited source precipitation during the final 72 h, anomalously warm seasonal conditions, relatively high surface velocities in part of

the source glacier and candidate seismic signals during the hours before the main event, without identifying a single immediate trigger. This evidential scope provides a process-rich reference against which an independent open-data reconstruction can be compared.

Here we present an uncertainty-aware reconstruction of the full Gyirong source–pathway–receptor cascade using open and publicly released Earth-observation data. The GF imagery, field photographs, seismic waveforms, video and autoRIFT/TICOI velocity results of Guo et al. (2026) were not used to define our measurements and served only as an independent comparison. This design permits cross-validation of first-order geometry and source classification while treating source-perimeter uncertainty, prespecified-window optical visibility limits, hydroclimatic product sensitivity, rapid-map damage grades and exposure intersections as distinct evidence classes.

We address three integrated questions: (1) what source geometry, propagation route and downstream consequences can be reconstructed and quantified while representing uncertainty; (2) what pre-event environmental and surface-observation conditions are supported, including hydroclimate, long-term regional context and the limits of routine optical detectability; and (3) what does the Gyirong event reveal about the capabilities and limits of open Earth observation for reconstructing rapidly evolving high-mountain hazard chains? The study does not perform formal event attribution or dynamic runout simulation, and pre-event observability is one component of the event reconstruction rather than its organizing question.

2 Study area and event

The mapped source is on the northern flank of the Langtang Lirung massif near the China–Nepal border, in steep glacierised terrain draining northward towards the Gyirong/Timure border corridor (Fig. 1). The preferred source centroid is 85.52701° E, 28.28858° N. Regional relief is high: the preferred source reaches 5221 m, whereas Gyirong Port lies near 1823 m. The upper route is steep and confined; farther downstream the drainage intersects settlements, roads, border facilities and the Bhote Koshi–Trishuli corridor. The area was also affected by snow and ice avalanches during the 2015 Gorkha earthquake, but the 2026 event was not classified here as earthquake-triggered (Kargel et al., 2016).

We use 26 August 2026 at 02:52:00 UTC (10:52 Beijing time) as the reference event time, corresponding to the reported mass-movement signal. An official account gives an approximate time near 10:30 Beijing time, and another signal time is 20 s later; these differences define the timing uncertainty (Jiangsu Earthquake Agency, 2026; China Geological Survey, 2026). All acquisition offsets and pre/post classifications use exact UTC timestamps rather than calendar dates.

Initial public descriptions variously used glacier collapse, ice avalanche, landslide, mudslide and flood. The China Geological Survey’s preliminary assessment described a high-elevation ice collapse that entrained moraine material and developed into a debris flow (China Geological Survey, 2026). We delineated the source independently before comparing it with these external descriptions. Place names are used descriptively and no boundary line is intended to express a jurisdictional position.

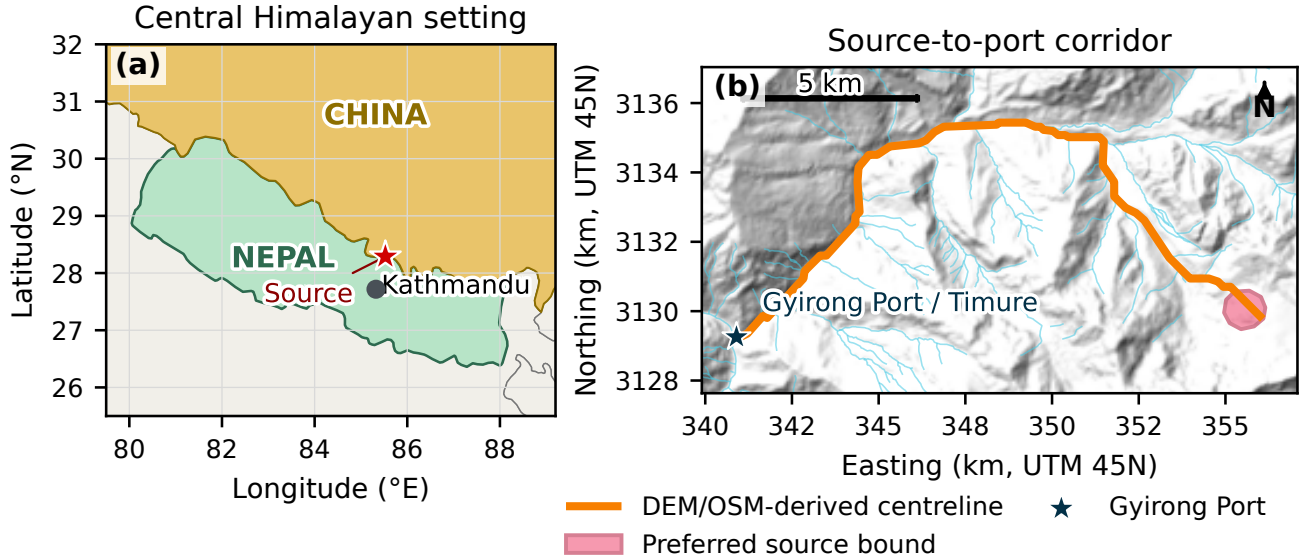


Figure 1. Regional setting of the 26 August 2026 Gyirong event. (a) Location near the China–Nepal border. (b) Preferred source bound and terrain-derived source-to-port centreline over Copernicus DEM hillshade. The 21.843 km centreline is a representative route constrained by drainage and post-event evidence, not a mapped inundation boundary. Data: Natural Earth country context, Copernicus DEM GLO-30, OpenStreetMap waterways, preferred source boundary and representative centreline. Place-name and boundary depiction is neutral.

3 Data and methods

3.1 Analysis scope, evidence classes and event clock

We reconstructed the event retrospectively using post-event multisensor observations to identify the source, route and downstream impacts. Source identification was based on convergent optical and radar evidence across plausible headwalls in the affected drainage; the reported source geometry was defined independently from earlier exploratory source-location attempts. To limit overinterpretation, we distinguish direct observations, derived measurements, supported process interpretations and unresolved questions throughout the analysis. Direct observations are visible features or features in independent mapping products; derived measurements are reproducible quantities calculated from defined data and geometries; supported interpretations connect multiple observations through process reasoning; and unresolved questions are not constrained by the available data. The accompanying reproducibility materials link the principal claims to their evidence and limitations.

The event time in Sect. 2 was applied before calculating all acquisition offsets. Images ending strictly before 02:52:00 UTC were pre-event; images starting afterward were post-event. No same-day acquisition was classified from date alone.

3.2 Satellite and ancillary data

Sentinel-2 Level-2A supplied 10 m bottom-of-atmosphere reflectance and scene-classification information (Drusch et al., 2012). Landsat-9 Collection 2 Level-2 supplied the first open post-event optical acquisition at 30 m sampling (Earth Resources Observation and Science Center, 2020). Public Planet disaster imagery supplied 3–4 m PlanetScope observations and footprint metadata under a CC BY-NC 4.0 licence (Planet Labs PBC, 2026).

Sentinel-1 radiometrically terrain-corrected backscatter supplied an event-bracketing amplitude-change layer; it was not processed as a full SLC, phase-interferometric or coherence stack (Torres et al., 2012). Copernicus DEM GLO-30 provided terrain elevation, slope, aspect and relief (European Space Agency and Copernicus Programme, 2021). OpenStreetMap waterways and roads helped constrain the representative corridor and exposure intersections (OpenStreetMap contributors, 2026). UNOSAT and Copernicus EMSR927 products were retained as independent rapid-response observations of the downstream footprint and damage grades, not as training labels or complete-census denominators (United Nations Satellite Centre, 2026; Copernicus Emergency Management Service, 2026). Table 1 summarizes the principal observations.

3.3 Source mapping and classification

The source search covered all plausible headwalls around the candidate drainage and did not begin from a scar-shaped region. Pre- and post-event optical images were compared on common grids. Mapping cues were a new fresh or scoured surface, disappearance or reorganisation of snow/ice, exposed darker rock or debris, headwall change and spatial continuity with the downstream affected corridor. Radar amplitude change was used as supporting convergence only because snow state, wetness and geometry can also change amplitude.

Three nested polygons express interpretive uncertainty in the perimeter of the mapped post-event changed/source surface. The conservative envelope contains only the fresh/scoured surface, snow/ice loss or reorganisation, and exposed rock/debris supported most consistently across visible observations. The preferred envelope represents the connected changed/source surface best supported by the combined observations. The inclusive envelope additionally contains plausible edges obscured or rendered ambiguous by cloud, haze, shadow and mixed surface change. These polygons do not represent measured failure depth, volume, rock/ice fractions or probabilistic confidence intervals, and they need not be identical to the initially detached body. The source geometry was finalized before comparison with external source locations. The source term was then selected from glacier detachment, hanging-glacier failure, ice avalanche, rock avalanche, mixed rock–ice slope failure or unresolved, with composition and initial kinematics treated separately.

3.4 Terrain and process-chain reconstruction

Source metrics were calculated in UTM zone 45N. Elevation, slope and aspect were sampled on a 10 m analysis grid derived from Copernicus DEM GLO-30. Aspect distribution used eight 45° sectors; mean aspect was circular. Local relief was the elevation range within a 500 m-radius neighbourhood. Maximum source dimension was the maximum planimetric separation within each polygon.

Table 1. Principal observations used in the reconstruction. Spatial sampling is nominal; locally usable coverage, timing and licence restrictions determine the effective evidence.

Source	Product or date	Sampling	Principal use and limitation
Sentinel-2	Level-2A, 2025–2026	10 m	Prespecified pre-event sequence and pre/post source change; affected by cloud, shadow, snow and illumination.
Landsat-9	Collection 2 Level-2, 26 Aug 04:47:48 UTC	30 m	First open post-event optical bracket, 1.93 h after failure; only 10.8 % of the source comparison area passed the validity mask.
PlanetScope	26 and 28 Aug	3.4–3.8 m	Partial high-resolution source support; source-covering clear fractions were 1.2–2.5 % and 6.1–11.3 %.
Sentinel-1	RTC amplitude, 16–28 Aug	approximately 10 m	Independent amplitude-change support; non-unique response, not InSAR, coherence or displacement.
Copernicus DEM	GLO-30	approximately 30 m	Source terrain and representative route geometry; a surface model with local registration limitations.
IMERG	V07 Late	0.1°, 30 min	Antecedent precipitation and matched 2001–2025 ranks; no source-scale rainfall.
ERA5-Land	Hourly	0.1°, 1 h	Regional precipitation, temperature and PDD context; grid elevation is 539 m below the source median.
UNOSAT/CEMS	Rapid maps, 26–28 Aug	product-dependent	Independent footprint and damage grades; preliminary and not a complete or field-validated census.

Planet imagery is used under its public disaster-data licence; redistribution requirements must be checked by the accountable authors.

A representative centreline connected the source to Gyirong Port while remaining within the independent UNOSAT affected footprint, except for a 15 m half-pixel support buffer. DEM and mapped waterways guided the centreline. Planar length was measured in UTM coordinates, surface length accumulated three-dimensional segment lengths, vertical drop used start and port endpoint elevations, and mean endpoint angle was $\tan^{-1}(H/L)$, where H is endpoint drop and L is planar route length. The profile retains raw 60 m DEM samples and a Savitzky–Golay smoothed curve over approximately 1.2 km. Local uphill steps in the raw record were considered DEM/centreline registration artefacts and were not interpreted dynamically. Process reaches were mapped as supported interpretations, not particle trajectories.

3.5 Pre-event optical observations

Six Sentinel-2 windows were selected before visual interpretation to represent approximately one year, six months, three months, one month, two weeks and the latest usable observation before failure. Within each target window, selection prioritized source-local scene-classification usability and then proximity to the nominal date; no date was chosen for apparent anomaly. All panels use a common projection, extent and comparable reflectance display. Interpretation was restricted to large-scale change resolvable at 10 m. We separately tested a feature-tracking configuration defined before inspecting the source results, but no source displacement estimate met both the stable-terrain-error and spatial-coherence requirements; displacement was therefore excluded from the main results.

3.6 Radar change

Sentinel-1 VV and VH radiometrically terrain-corrected amplitude observations bracketing the event were co-registered on the event grid and compared in logarithmic space. We interpreted only spatially coincident source or corridor changes supported by optical or rapid-map evidence. Conventional InSAR was not attempted because steep topography, layover/shadow, snow/ice decorrelation and potentially non-phase-resolvable displacement made a robust phase result unlikely. Coherence and amplitude-change methods can assist rapid landslide mapping, but are non-unique and must be checked against independent information (Burrows et al., 2020; Jacquemart and Tiampo, 2021).

3.7 Hydroclimatic context

ERA5-Land hourly 2 m temperature and total precipitation and GPM IMERG V07 Late half-hourly precipitation were obtained from their official archives (Copernicus Climate Change Service, 2019; Muñoz-Sabater et al., 2021; Huffman et al., 2023, 2020). Only the five prespecified antecedent windows were evaluated: 24 h, 3 d, 7 d, 14 d and 30 d. The ERA5-Land cut-off was the complete hour ending at 02:00 UTC (52 min before failure); the IMERG cutoff was the complete half-hour ending at 02:30 UTC (22 min before failure). Event values were compared with matched 26 August windows for 2001–2025 ($n = 25$), excluding 2026.

The empirical percentile was $100(n_{<} + 0.5n_{=})/n$, where $n_{<}$ and $n_{=}$ are historical values below and equal to the event value. Robust anomalies used the median and median absolute deviation; no formal independent-window p values were assigned because the accumulation windows overlap. Daily ERA5-Land temperature statistics used complete UTC days. A warm spell was defined before analysis as consecutive complete days ending 25 August on which daily mean temperature exceeded the matched day-specific 90th percentile. Positive degree days (PDD) were $\sum \max(T_{\text{daily}}, 0)$ for complete days. PDD is a temperature-index proxy and is not a direct measurement of melt (Wake and Marshall, 2015).

The primary temperature and PDD values were not lapse-rate corrected. The ERA5-Land grid elevation (approximately 4452 m) is 539 m below the preferred source median (4991 m). Spatial sensitivity was evaluated with the source cell, neighbouring cells and a predefined 3×3 support. Both products have approximately 0.1° spatial sampling and cannot resolve

headwall-scale precipitation, local convection, source-scale snowfall, subsurface water, meltwater routing, permafrost state or failure-plane hydrology.

3.8 Historical environmental and exposure context

Longer-term July–August temperature was evaluated using the same predefined spatial supports and seasonal definition. Hourly ERA5-Land data for 1951–2025 were reduced to annual July–August means at the source-nearest grid cell and over the fixed Central Himalaya regional support. We calculated ordinary-least-squares (OLS) trends with 95 % confidence intervals, Theil–Sen slopes, Mann–Kendall statistics, and 10- and 20-year running means for 1951–2025, 1976–2025 and 2001–2025. The extended record was validated by reproducing the previously calculated 2001–2025 trends before analysis of the longer time series (Muñoz-Sabater et al., 2021).

Berkeley Earth monthly land temperature over the same regional box, where available, supplied an independent July–August sensitivity record for 1951–2024 (Rohde et al., 2013; Rohde and Hausfather, 2020). CMIP6/DAMIP forcing consistency used four prespecified models with paired historical/all-forcing and historical-natural simulations, one member per model and equal model weight, over the same regional box. Comparisons were restricted to July–August 1951–2020 and 1971–2020 because the observational record used here is seasonal rather than full-year. This comparison tests whether the observed trend lies more consistently within the small all-forcing or natural-only model distributions; it is not optimal fingerprinting, formal event attribution or an estimate of the anthropogenic fraction of warming (Gillett et al., 2016).

Six historical source-image epochs (1972, 1987, 2000, 2008, 2020 and 2026) were inspected within a fixed source-context region. Cross-sensor interpretation was harmonised to a 120 m minimum mapping unit. Because seasonal snow, terrain shadow and sensor differences prevented like-for-like delineation, we did not estimate a quantitative source-cryosphere trend. We also examined the images for direct surface disturbance within fixed 0.5, 1, 2 and 5 km source buffers. Historical built-up exposure was measured independently with GHSL R2023A built-up-surface grids for 1975, 1990, 2000, 2015 and 2020 in four fixed zones: the UNOSAT affected footprint, Timure/Gyirong Port, Syapru Besi and a fixed 500 m corridor buffer. Built-up area and fraction were calculated at nominal zone geometry and with low/high boundary-sampling sensitivity; they measure exposure, not population, damage, casualties or hazard magnitude (Pesaresi et al., 2024).

3.9 Channel change, damage and exposure

Twelve cross-valley transects were placed at terrain- and channel-defined positions before widths were inspected. At each transect we separately attempted to measure a channel-connected pre/post water segment and the width of the independent mapped disturbed footprint. A water width was reported only when both dates met the prespecified valid-pixel and connectivity rule. Disturbed width therefore does not imply flowing-water width. Its reported uncertainty contains a ± 20 m pixel-edge component but not a formal interpreter-boundary term.

Copernicus damage grades were summarized by delivered feature class and area of interest. Road lengths were calculated from mapped line geometry; buildings, transport points and facilities were counted as mapped features. OpenStreetMap roads

and bridge-tagged features were intersected with the UNOSAT footprint to characterize exposure only. No casualty or population estimate was inferred, and rapid-map counts were not treated as complete inventories.

4 Results

4.1 Source identification and geometry

The final pre-event Sentinel-2 image was acquired at 05:00:49 UTC on 24 August, 45.85 h before the failure clock. Post-event Landsat-9, Sentinel-2 and partly unobscured PlanetScope imagery showed a new fresh or scoured surface, loss or reorganisation of snow and ice, and exposure of darker debris or rock in the same high-elevation slope sector (Fig. 2). Sentinel-1 RTC amplitude change between 16 and 28 August was spatially coincident with the optical change, although the radar response was not unique to mass movement. The Pelican and SkySat strips examined did not cover the mapped source; source-covering PlanetScope scenes were strongly affected by cloud and haze.

The preferred mapped post-event changed/source surface was 1.009 km², bracketed by conservative and inclusive envelopes of 0.491 and 1.841 km² (Fig. 3; Table 2). Within the preferred envelope, Copernicus DEM elevations ranged from 4355 to 5221 m, with median and mean elevations of 4991 and 4940 m. Median and mean slopes were 34.8 and 34.1°. The circular mean aspect was 280.8°; west- and northwest-facing pixels comprised 48.2 % and 33.8 % of the mapped area. Median local relief within a 500 m radius was 840 m, and the maximum planimetric source dimension was 1151 m.

Direct observations establish change involving both snow/ice-covered and exposed rock/debris surfaces. Combined with the steep source terrain, this supports the term *mixed rock-ice slope failure*. The data do not resolve whether rupture initiated in bedrock, glacier ice or their interface, nor the rock-to-ice proportion, failure depth or initial kinematics. No volume was calculated because no suitable co-registered pre/post DEM pair was available.

4.2 Source-to-downstream cascade

A representative centreline constrained by the independent UNOSAT footprint and mapped drainage connected the source to Gyirong Port/Timure (Fig. 4). The planar source-to-port length was 21.843 km and the DEM-derived surface length was 22.710 km. The route descended from approximately 5220 m at the source crown to 1823 m at the port, an endpoint drop of 3396 m and a mean endpoint angle of 8.84° (Fig. 5). The adopted topographic transition from the steep initial avalanche reach, where entrainment may have occurred, to channelised transport was placed at the first sustained descent below 3200 m, 5.94 km from the source. This divider is interpretive rather than a directly observed flow transition.

The mapped sequence comprised the source polygon, an interpreted initial avalanche reach in which entrainment may have occurred, channelised transport to the port and a downstream mudflow/flood route. The independent UNOSAT product mapped a 37.353 km² combined mudflow/rockflow footprint. Copernicus mapped 1.259 km² and 1.111 km² event polygons in its Timure/Gyirong Port and Syapru Besi areas, respectively. The available observations did not resolve a separate short-lived blockage or breach geometry.

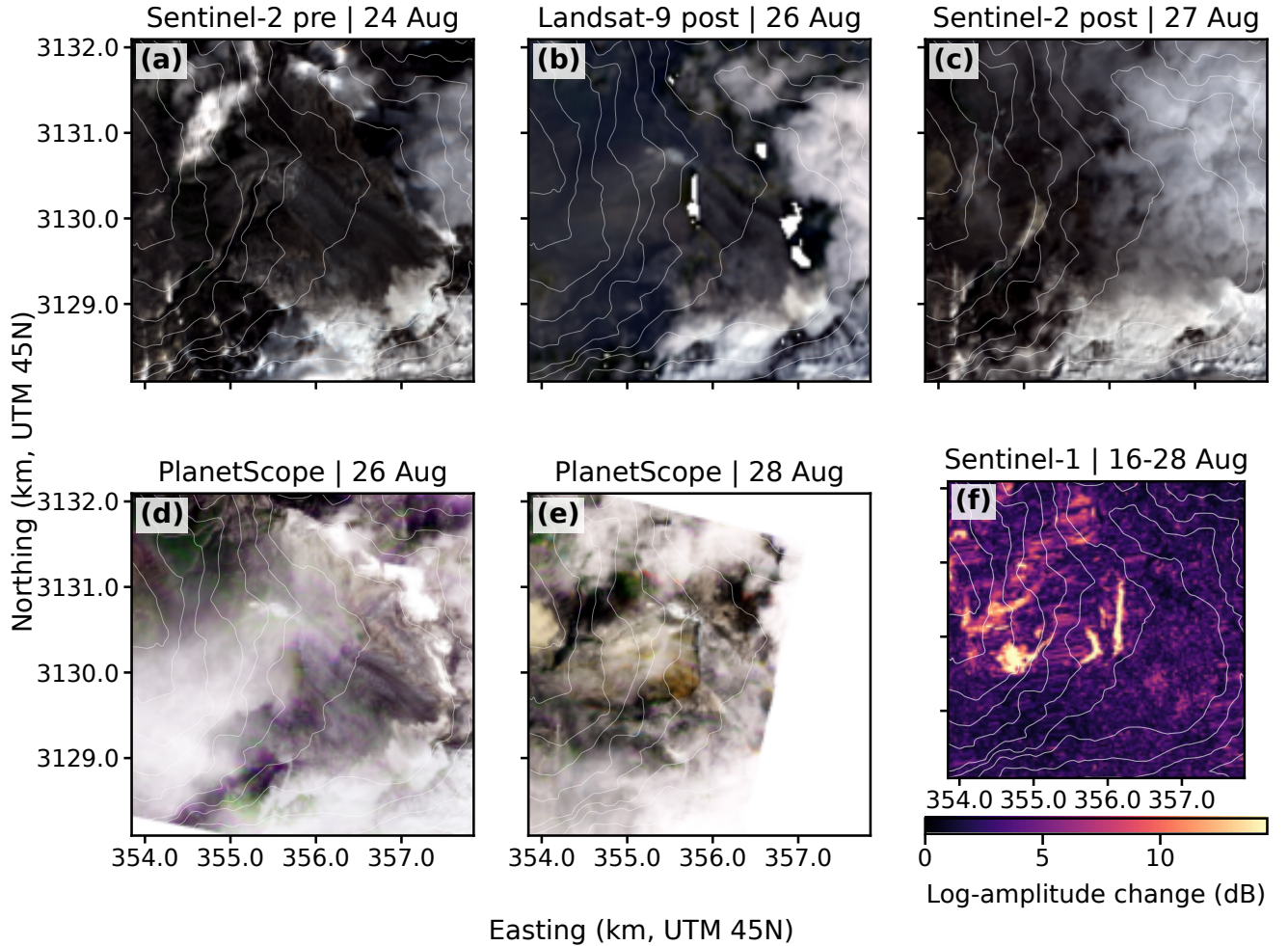


Figure 2. Common-grid source-region evidence comparison without external source labels. Acquisition times (UTC) were Sentinel-2 pre-event, 24 August 05:11; Landsat-9, 26 August 04:47; Sentinel-2, 27 August 05:10; PlanetScope, 26 August 05:44 and 28 August 04:57; and Sentinel-1 RTC, 16 and 28 August 12:21. The observations show convergent but variably cloud-limited surface change. White or obscured areas are not interpreted, and radar amplitude change alone is not diagnostic. Data: Sentinel-2 Level-2A, Landsat Collection 2 Level-2, PlanetScope public disaster imagery and Sentinel-1 RTC.

4.3 Pre-event surface observations and observational limits

The six prespecified Sentinel-2 windows were 6 October 2025, 13 February 2026, 16 May 2026, 2 July 2026, 12 August 2026 and 24 August 2026 (Fig. 6). Source-region usable fractions varied from 42 % to 100 % because of cloud, shadow, seasonal snow and illumination. This routine Sentinel-2 sequence showed no unambiguous large-scale pre-failure surface change at

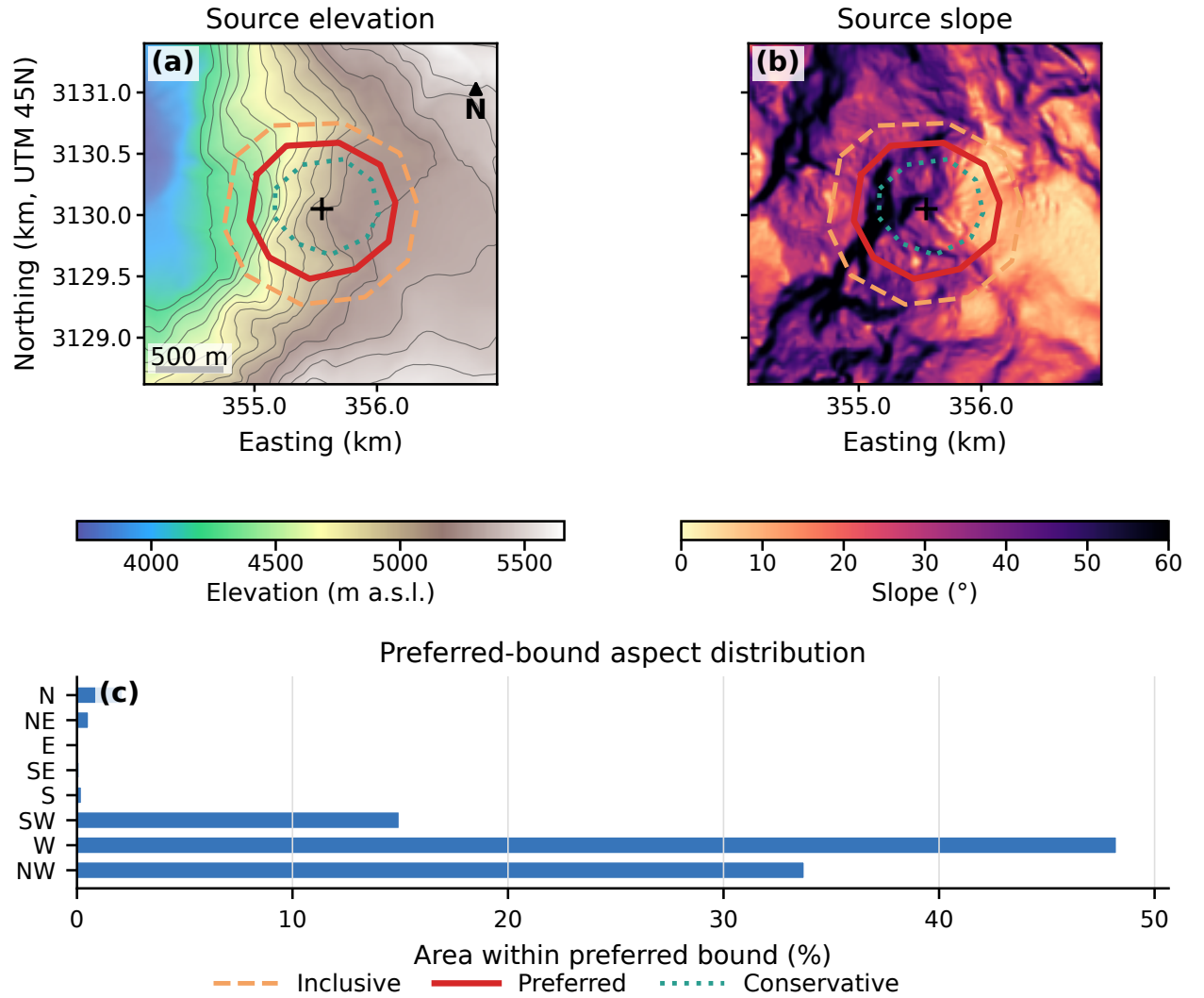


Figure 3. Mapped changed/source-surface geometry and terrain uncertainty. (a) Conservative, preferred and inclusive mapping envelopes over elevation. (b) DEM-derived slope. (c) Aspect distribution within the preferred envelope. The nested polygons describe perimeter interpretation, not the volume or material fractions of the initially detached body and not probabilistic confidence intervals. No source volume is estimated. Data: multisensor source boundaries and Copernicus DEM GLO-30.

10 m sampling, and the final usable optical observation preceded failure by 45.85 h. This result does not exclude persistent motion detectable through dense feature tracking, sub-pixel or obscured changes, or activity during the final observation gap.

The first open post-event optical acquisition was Landsat-9 at 04:47:48 UTC, 1.93 h after failure, but only 10.8 % of the source comparison area passed the validity mask. PlanetScope first covered the source at 05:44:56 UTC, 2.88 h after failure;

Table 2. Mapped changed/source-surface and route geometry. Conservative, preferred and inclusive envelopes express perimeter interpretation and are not confidence intervals or detached-volume estimates.

Metric	Conservative	Preferred	Inclusive	Qualification
Changed/source-surface area (km ²)	0.491	1.009	1.841	Nested multisensor mapping envelopes
Elevation range (m)	4543–5200	4355–5221	4261–5261	Copernicus DEM GLO-30
Median elevation (m)	5045	4991	4977	DEM-derived
Median slope (°)	35.7	34.8	32.1	DEM-derived
Circular mean aspect (°)	280.5	280.8	285.2	Clockwise from north
Median local relief (m)	804	840	849	500 m-radius neighbourhood
Maximum source dimension (m)	798	1151	1526	Maximum planimetric separation
Planar route to port (km)	–	21.843	–	Representative centreline, not a particle trajectory
Surface route to port (km)	–	22.710	–	DEM-derived three-dimensional length
Endpoint drop (m)	–	3396	–	Approx. 5220 to 1823 m
Mean endpoint angle (°)	–	8.84	–	First-order geometric descriptor

No source volume is reported because no suitable co-registered pre/post DEM pair exists.

clear fractions within covered pixels were only 1.2–2.5 %. Sentinel-2 at 05:10:45 UTC on 27 August provided 37.5 % locally usable source coverage, and PlanetScope on 28 August provided 6.1–11.3 % clear coverage. The acquisitions jointly bracket the source change but do not provide one uniformly clear high-resolution post-event perimeter. No displacement estimate met the predefined stable-terrain-error and spatial-coherence criteria.

4.4 Short- and long-term environmental context

All five precipitation windows were complete to the final product-specific cutoff. IMERG recorded 4.0 and 8.3 mm over 24 h and 3 d, at the 20th and 4th percentiles of matched 2001–2025 periods. Its 7, 14 and 30 d totals were 16.8, 44.6 and 103.4 mm, each lower than all 25 matched historical years. ERA5-Land totals were 5.7, 20.2, 74.5, 194.0 and 422.0 mm over 24 h, 3 d, 7 d, 14 d and 30 d, at the 16th, 16th, 44th, 64th and 68th percentiles, respectively (Fig. 7; Table 3).

The products disagreed substantially in amount and historical rank, especially over 7–30 d. Their common result was the absence of an unusually wet gridded antecedent period. This is not evidence that the source was dry and does not exclude unresolved local precipitation.

Regional thermal conditions were anomalously warm. The ERA5-Land 7 d mean 2 m temperature was 9.43 °C, above all 25 matched years; the 14 d mean was 9.36 °C, at the 96th percentile. Daily mean temperature exceeded the matched daily 90th-percentile threshold for five consecutive complete days ending 25 August.

Unadjusted PDD accumulation was 65.94 °C d over 7 d, above all 25 matched historical years. Totals over 14 d, 30 d and 1 July–25 August were 131.11, 279.44 and 526.02 °C d, at the 96th, 88th and 96th percentiles. The 539 m elevation mismatch

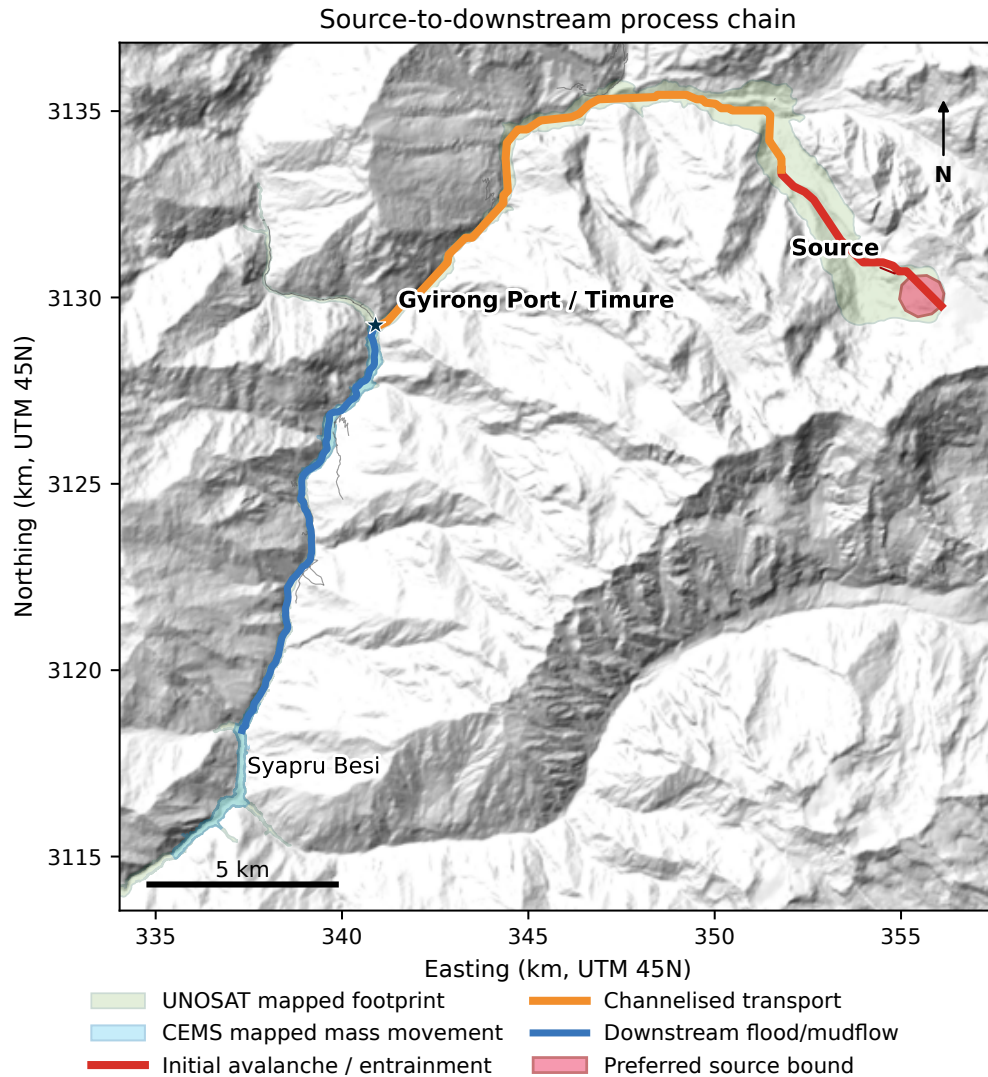


Figure 4. Reconstructed source-to-downstream process chain. The preferred source connects to a steep initial avalanche reach, channelised transport to Gyirong Port/Timure, and a downstream mudflow/flood route towards Syapru Besi. Entrainment within the upper reach is plausible but is not independently quantified by these data. Process lines and reach boundaries are interpretations supported by terrain and mapped change; shaded UNOSAT and Copernicus EMS polygons are independent rapid-response observations. Data: Copernicus DEM GLO-30, UNOSAT, Copernicus EMSR927, OpenStreetMap and the event GIS dataset.

and 0.1° grid spacing mean that these values describe regional melt-season thermal conditions, not source temperature or source-ice melt.

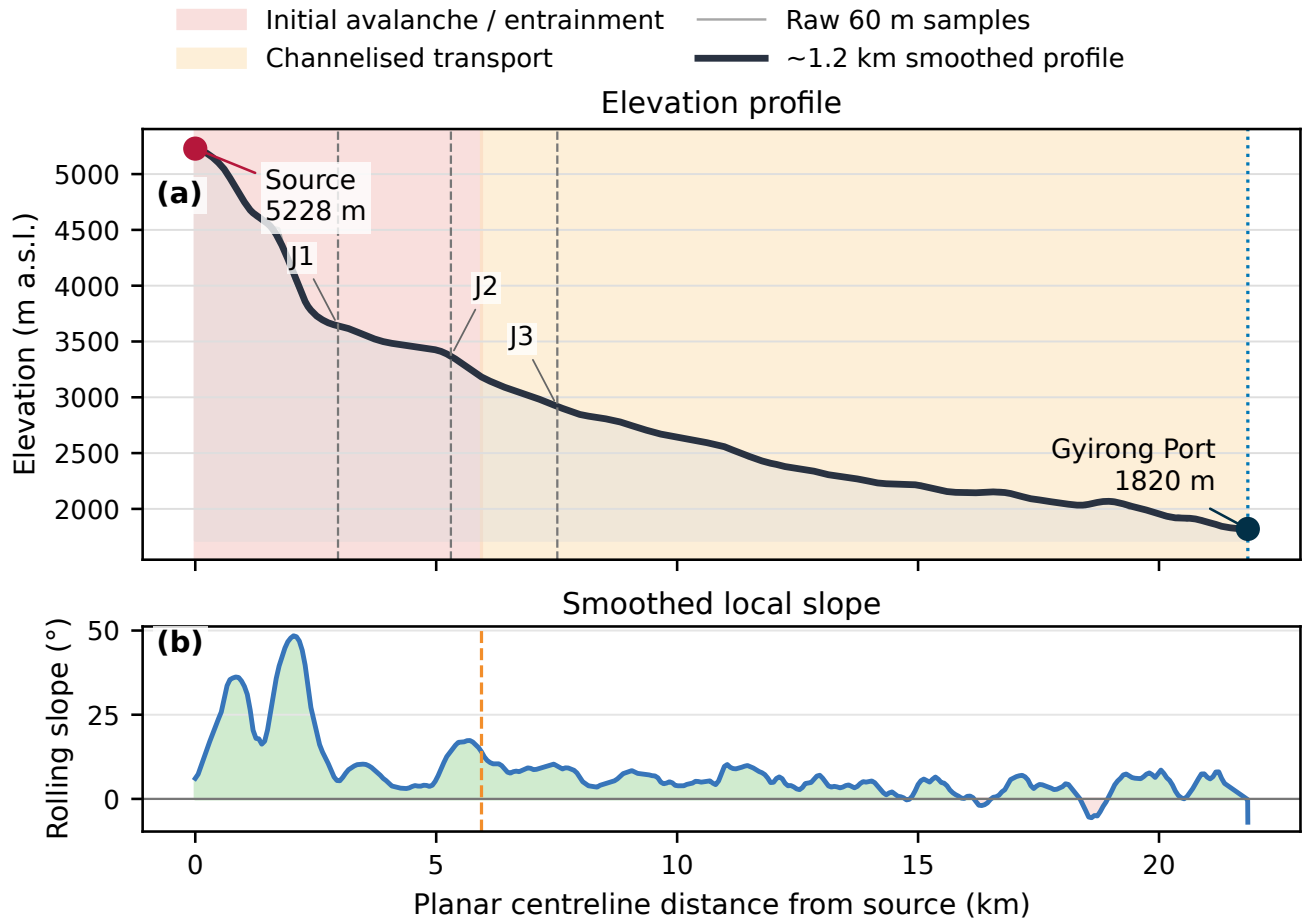


Figure 5. Longitudinal source-to-port terrain profile. Raw 60 m samples are shown with a Savitzky–Golay smoothed elevation and rolling-slope representation. The route descends approximately 3396 m over a 21.843 km planar distance. Shading denotes supported process reaches; J1–J3 mark major mapped drainage junctions. Data: Copernicus DEM GLO-30, representative centreline and OpenStreetMap waterways. The displayed elevation is smoothed over approximately 1.2 km; local raw-profile uphill steps are registration artefacts and are not flow behaviour.

Regional July–August ERA5–Land temperature increased by $0.1208\text{ }^{\circ}\text{C decade}^{-1}$ during 1951–2025 (95 % OLS confidence interval $0.0894\text{--}0.1521\text{ }^{\circ}\text{C decade}^{-1}$; Theil–Sen slope $0.1203\text{ }^{\circ}\text{C decade}^{-1}$; Mann–Kendall $p = 1.47 \times 10^{-9}$). The corresponding source-nearest grid-cell trend was $0.0666\text{ }^{\circ}\text{C decade}^{-1}$. For 1976–2025, the regional and source-nearest trends were 0.1482 and $0.0675\text{ }^{\circ}\text{C decade}^{-1}$, respectively. These differences show that the regional mean cannot be treated as a source-temperature measurement (Fig. 8a).

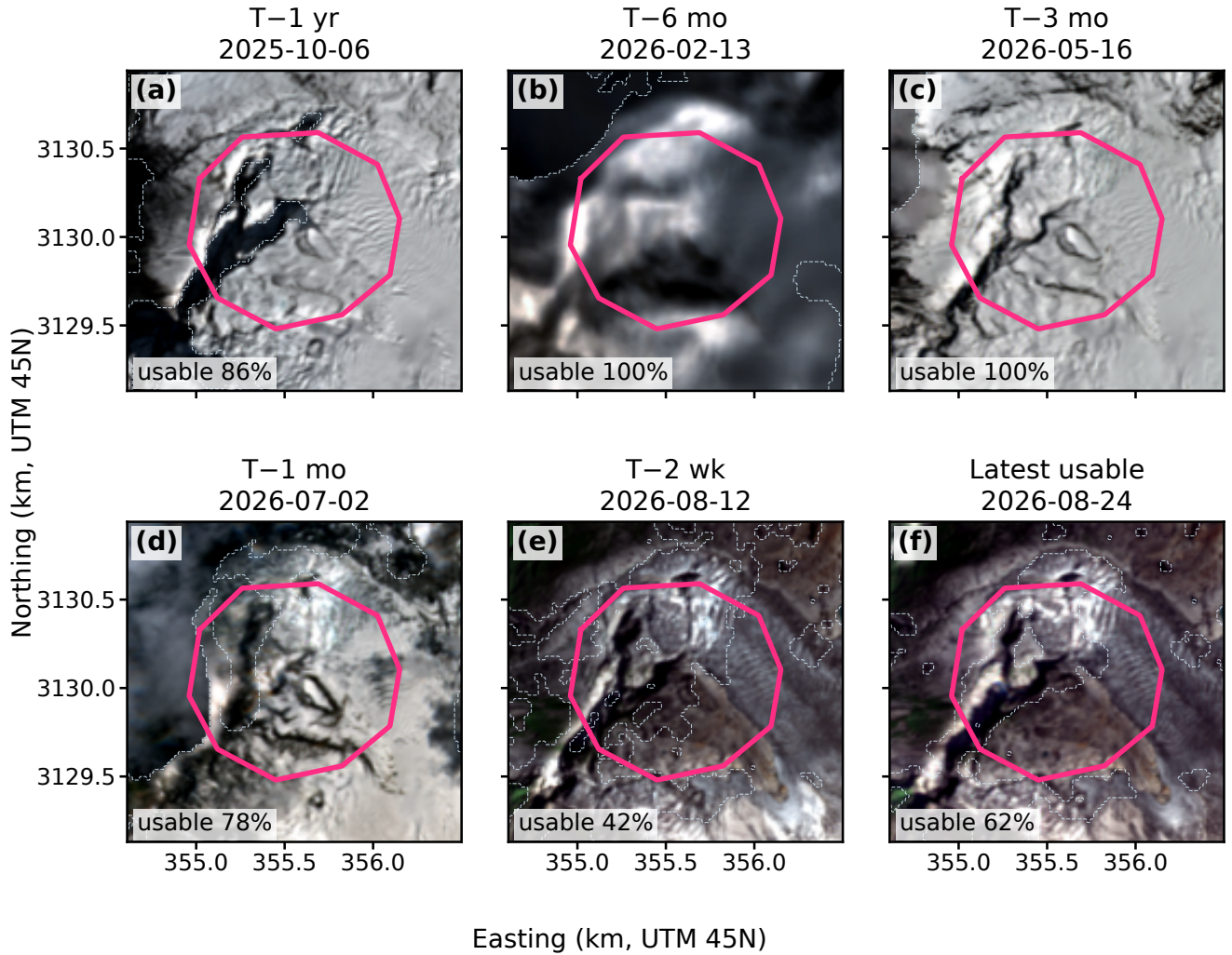


Figure 6. Sentinel-2 source-region sequence for six prespecified pre-event windows using a common extent, projection and display. The sequence shows no unambiguous large-scale pre-failure surface change at 10 m sampling. Cloud, shadow and seasonal snow vary among panels; the latest usable observation preceded failure by 45.85 h. This does not exclude persistent motion, sub-pixel or obscured changes, or activity during the final observation gap. Data: Sentinel-2 Level-2A and the preferred source boundary.

Berkeley Earth independently gave a positive 1951–2024 July–August regional trend of $0.1531\text{ }^{\circ}\text{C decade}^{-1}$ (95 % confidence interval 0.1128–0.1933), compared with $0.1156\text{ }^{\circ}\text{C decade}^{-1}$ (0.0840–0.1472) from ERA5-Land over the directly comparable period. For 1951–2020, the observed ERA5-Land trend was $0.0966\text{ }^{\circ}\text{C decade}^{-1}$, within the four-model all-forcing range (0.052–0.702) and outside the natural-only range (−0.052 to −0.005). For 1971–2020, the observed $0.1252\text{ }^{\circ}\text{C decade}^{-1}$ trend again lay inside the all-forcing range (0.061–0.687) and outside the natural-only range (−0.031–0.053; Fig. 8b). This small-ensemble result establishes forcing consistency, not event attribution.

Table 3. Hydroclimatic conditions preceding failure. Historical ranks compare matched 26 August periods in 2001–2025 ($n = 25$). Values describe gridded regional context and do not classify a trigger.

Variable	Window	IMERG	ERA5-Land	Historical rank
Precipitation (mm)	24 h	4.0	5.7	20th; 16th percentile
	3 d	8.3	20.2	4th; 16th percentile
	7 d	16.8	74.5	lower than all 25; 44th percentile
	14 d	44.6	194.0	lower than all 25; 64th percentile
	30 d	103.4	422.0	lower than all 25; 68th percentile
Mean 2 m temperature (°C)	24 h	–	9.41	92nd percentile
	3 d	–	9.59	above all 25
	7 d	–	9.43	above all 25
	14 d	–	9.36	96th percentile
	30 d	–	9.31	88th percentile
PDD (°C d)	7 d	–	65.94	above all 25
	14 d	–	131.11	96th percentile
	30 d	–	279.44	88th percentile
	1 Jul–25 Aug	–	526.02	96th percentile

IMERG and ERA5-Land have approximately 0.1° sampling. The ERA5-Land grid elevation is approximately 4452 m, 539 m below the preferred source median. Temperature and PDD are not lapse-rate corrected; PDD is not a direct melt measurement.

4.5 Geomorphic consequences

The independent UNOSAT footprint intersected all 12 terrain-positioned transects (Supplement). Disturbed-corridor widths ranged from 145 to 1298 m, with a median of 533 m. The nine source-to-port transects had a median width of 970 m, whereas the three downstream transects had a median width of 195 m. The ± 20 m pixel-edge component does not include formal uncertainty in the independently mapped boundary.

No transect yielded a defensible pre/post water-width pair under the prespecified valid-pixel and channel-connectivity rule. The 24 August baseline was obscured at several lower-valley transects, and elsewhere the automated masks did not resolve a channel-connected water segment above the sensor-specific minimum width. Water-width change is therefore not measured, rather than zero. Available imagery likewise did not support consistent separation of scour, entrainment, transport, deposition, new/lost water and temporary blockage into distinct areal classes.

The transect map and complete width profile are presented in the Supplement because they characterize observation limits rather than provide a direct measurement of water-width change.

Table 4. Independent rapid-map damage grades and exposure intersections. Feature counts and lengths describe the published product layers, not complete inventories.

Metric	Value	Evidence type	Qualification
UNOSAT affected footprint	37.353 km ²	Rapid-map observation	Preliminary; not field validated
Destroyed building points	695	CEMS damage grade	Mapped points, not a building census
Damaged building points	65	CEMS damage grade	Mapped points
Possibly damaged building points	104	CEMS damage grade	Mapped points
Destroyed road length	15.671 km	CEMS damage grade	Mapped lines, not network completeness
Possibly damaged road length	1.661 km	CEMS damage grade	Mapped lines
Destroyed transport/facility features	6 / 2	CEMS damage grade	Points / polygons in the mapped products
OSM road exposure intersection	7.414 km	Exposure	Not observed damage; completeness unknown
OSM bridge-tagged intersections	2	Exposure	Not observed damage; inventory incomplete

No casualties or population exposure are inferred.

4.6 Infrastructure impacts and historical exposure

Copernicus EMSR927 layers contained 433 affected building points in AOI01 (Syapru Besi) and 431 in AOI02 (Timure/Gyirong Port). Across both areas, 695 building points were graded destroyed, 65 damaged and 104 possibly damaged. The same products mapped 15.671 km of destroyed road, 1.661 km of possibly damaged road and 3.872 km of road with no visible damage in the published affected-feature layers. Six transport points and two facility polygons were graded destroyed (Fig. 9; Table 4).

OpenStreetMap overlay identified 7.414 km of mapped road, represented by 25 line features, and two bridge-tagged features intersecting the UNOSAT footprint. These are exposure intersections, not observations of damage. OpenStreetMap segmentation and completeness affect both counts, and no building-population exposure or casualty estimate was produced.

GHSL built-up surface increased between 1975 and 2020 in every fixed downstream zone (Fig. 8c). Within the UNOSAT affected footprint, mapped built-up area increased from 0.0359 to 0.4412 km²; Timure/Gyirong Port increased from 0.00541 to 0.0363 km², Syapru Besi from 0.00258 to 0.0246 km², and the fixed corridor from 0.000728 to 0.00952 km². In each zone, the 2020 lower boundary-sensitivity estimate exceeded the 1975 upper estimate. These changes quantify growth of built-up exposure only; they do not measure damage, population growth or hazard amplification.

5 Discussion

5.1 Source classification and uncertainty

Our strongest source evidence is the convergence of a new optical surface change, snow/ice loss or reorganisation, exposed darker rock/debris, steep topography and spatially coincident Sentinel-1 amplitude change. This combination makes *mixed rock–ice slope failure* more defensible than either glacier collapse or rock avalanche alone. Independent GF imagery and field

photographs presented by Guo et al. (2026) show glacier ice and rock involvement and residual hanging ice, providing external consistency with this material classification. Those observations were not used to define our source geometry.

The two studies attach different areas to the source. Our preferred 1.009 km^2 value, bounded by conservative and inclusive mapping envelopes of $0.491\text{--}1.841 \text{ km}^2$, describes the connected post-event changed/source surface visible across our optical and radar evidence. Guo et al. (2026) report an approximately 0.72 km^2 collapsed ice–rock body using GF and other event-specific observations. The estimates should not be treated as interchangeable or statistically combined: one is explicitly a changed-surface perimeter with obscured-edge envelopes, whereas the other is described as the collapsed source body. Differences in sensor resolution, observation dates, cloud and haze, inclusion of fresh/scoured margins, and manual perimeter interpretation are plausible non-exclusive explanations. Without both digital polygons on a common grid, their overlap and boundary displacement cannot be quantified.

The classification remains interpretive. Optical appearance cannot quantify rock-to-ice proportion, and the lack of a co-registered pre/post DEM prevents failure-depth and volume estimates. The source envelopes express visible perimeter ambiguity, not volumetric uncertainty. The mapped geometry also differs from the low-angle, soft-bed detachment settings documented at Aru and Flat Creek (Kääb et al., 2018; Jacquemart et al., 2020; Kääb et al., 2021). Because neither our data nor the external consistency establishes whether rupture began in bedrock, glacier ice or their interface, a more specific initiation label would overstate the evidence.

No mapped evidence of large persistent direct anthropogenic surface disturbance was identified at the source within the observable historical record, subject to the stated resolution and visibility limits. This bounded non-detection is not proof that no human activity occurred: narrow, transient, subsurface, unmapped or terrain-shadowed disturbance could remain unresolved in the 1972–2026 images.

5.2 Cascade transformation and mobility

Continuity from the source through a steep confined valley to the independently mapped downstream footprint supports a channel-coupled cascade in which the initial mass fragmented and entered channelised transport. Our independent 21.843 km plan length, 3396 m endpoint drop and 8.84° endpoint angle constrain the source-to-port geometry. They do not quantify rheology, velocity, discharge, entrained volume or the trajectory of individual particles. The terrain-based transition at 5.94 km is likewise an interpretive reach boundary.

Event-specific evidence for progressive erosion and entrainment is stronger in Guo et al. (2026), who combine GF imagery, field observations and process information to infer bed and bank scour, incorporation of unconsolidated material, and interaction with channel water and sediment. Our mapped corridor is consistent with such amplification, as are analogues from Kolka, Chamoli and Sedongpu (Huggel et al., 2005; Shugar et al., 2021; Kääb and Girod, 2023; Zhang et al., 2025), but our data neither isolate erosion and deposition volumes nor demonstrate how much mass was added. Accordingly, progressive entrainment is external event-specific support for the cascade interpretation, not an independently quantified result of this study. A dynamic model would add false precision without source volume, material fractions, hydrograph or suitable validation.

5.3 Environmental context and limits of causal inference

Our prespecified 3 d IMERG Late accumulation of 8.3 mm is numerically consistent with the approximately 8.3 mm source-area accumulation over the final 72 h reported independently by Guo et al. (2026). This agreement supports limited immediate antecedent rainfall in that gridded product. It does not establish source-scale rainfall, because product version, cut-off convention, grid-cell selection and complex-orographic sampling still require care.

The broader five-window comparison adds information beyond that common 72 h value. Neither IMERG nor ERA5-Land placed any of the five prespecified event windows in an unusually wet upper tail, but the products disagreed strongly in amount and historical rank over 7–30 d. ERA5-Land indicated anomalously warm regional conditions: the 3 and 7 d mean temperatures and 7 d PDD ranked above all 25 matched 2001–2025 years, and an objectively defined five-day warm spell ended on 25 August. This is consistent with the warmer spring and summer conditions reported by Guo et al. (2026), while our contribution is the matched-year ranking, prespecified windows, two-product comparison and explicit representativeness analysis.

The warm regional period is physically compatible with enhanced seasonal melt or thermal/hydrological preconditioning in ice-bearing terrain, but such preconditioning is not demonstrated here. The approximately 0.1° products smooth complex orography, and the ERA5-Land grid elevation is 539 m below the preferred source median. PDD is a temperature-index proxy, not an energy-balance melt observation (Wake and Marshall, 2015). No observation constrains source-scale meltwater production, water storage or routing, pore pressure, ice temperature, permafrost distribution, fracture ice or failure-plane hydrology. Permafrost conditions are particularly poorly known across the Hindu Kush Himalaya (Gruber et al., 2017). Consequently, neither our analysis nor the external comparison proves or disproves a thermal, melt or rainfall trigger.

Regional July–August warming is supported by ERA5-Land and Berkeley Earth and is more consistent with CMIP6 simulations including anthropogenic forcing than with natural-only simulations. The agreement is strongest for the sign and multi-decadal scale of regional summer warming; model spread, sparse topographic sampling and the seasonal observational target preclude a more specific attribution statement. This regional attribution does not establish that anthropogenic warming caused or mechanically contributed to the 2026 rupture. The source-specific links involving permafrost, meltwater, pore pressure, fracture weakening, ice or rock temperature and failure-plane hydrology remain unresolved. The independently observed short warm period is therefore regional environmental context, not evidence of a human-induced trigger or source-scale causal mechanism (Jiang and Zhou, 2023).

5.4 Pre-event observability across time scales

The routine Sentinel-2 sequence showed no unambiguous large-scale pre-failure surface change, and the final usable optical observation preceded failure by 45.85 h. No corrected source-displacement estimate met the predefined stable-terrain-error and spatial-coherence criteria. These findings apply only to resolvable surface change in the six prespecified optical snapshots. They do not exclude persistent motion detectable through dense feature tracking, sub-pixel change, change obscured by cloud, shadow or snow, or activity within the final observation gap (Lacroix et al., 2018; Chudley et al., 2022; Zheng et al., 2023; Altena et al., 2022).

The dedicated analysis of Guo et al. (2026) addresses a different observable. Their supplement describes autoRIFT displacement measurements from 211 Sentinel-2 scenes acquired between January 2024 and August 2026, 60–400 d image-pair baselines, stable-area correction, robust TICOI inversion and 15 d velocity output. They report relatively high 2024–2026 velocity in parts of the source glacier. A multi-image velocity inversion can recover persistent motion that is not visible as an unambiguous morphological change in six snapshots; our snapshot-based non-detection therefore neither reproduces nor refutes their velocity result. Reviews of avalanching-glacier instability likewise caution that velocity patterns are process-regime dependent and are not deterministic indicators by themselves (Faillettaz et al., 2015; Kääb et al., 2021).

Guo et al. additionally identify several candidate KKN seismic signals during the approximately 3 h before the principal event and relate some timing to eyewitness or video observations, while declining to interpret them as deterministic indicators (Guo et al., 2026). Those signals fall well inside our 45.85 h optical observation gap. The combined record therefore illustrates an observation-scale hierarchy: multi-year velocity and seasonal climate describe long-term state; days-to-weeks optical snapshots and thermal anomalies describe intermediate conditions; seismic or video records can sample activity over hours; and the main cascade propagates over minutes. Agreement or non-detection at one scale cannot be transferred automatically to another.

The source’s steep geometry and snow/ice cover also weaken a conventional InSAR interpretation. Layover, shadow, decorrelation and phase ambiguity may suppress or distort deformation, while amplitude change is non-unique. We therefore retain Sentinel-1 only as post-event amplitude-change support and do not claim a radar-derived deformation history.

5.5 Downstream consequences and changing exposure

The observed consequence extended far beyond the source. Independent rapid maps show a broad disturbed corridor and mapped damage at Timure/Gyirong Port and Syapru Besi. The 695 building points and 15.671 km of road graded destroyed summarize the published Copernicus EMS products, but their denominator, omission rate and field accuracy are unknown. The UNOSAT polygon and transect intersections describe a mapped affected corridor, not instantaneous flow width, water width, deposition area or inundation depth.

This impact emphasis is complementary to the process-amplification focus of Guo et al. (2026). OpenStreetMap intersections answer which mapped assets lie within the rapid-map footprint, not which assets were damaged. Separating imagery-based change mapped in this study, independent rapid-map damage grades and exposure intersections avoids converting incomplete inventories into loss rates or casualty estimates. Water-width change could not be measured because cloud-obscured baselines and mixed pixels prevented channel-connected segments from meeting the prespecified criteria; disturbed widths therefore cannot be recast as flood-water widths.

Historical GHSL data independently show substantial growth of downstream built-up exposure. This represents anthropogenic amplification of exposure, not amplification of hazard magnitude or of the source failure itself. Built-up surface is not a complete measure of population, occupancy, asset value or vulnerability; it demonstrates only that more mapped development was situated within the fixed downstream zones by 2020 (Pesaresi et al., 2024).

5.6 What open Earth observation can and cannot reconstruct

The comparison shows that observing systems should be combined without being conflated. High-resolution forensic imagery and field photographs constrain source composition and erosion evidence; multi-year feature tracking constrains glacier motion; seismic, infrasound or video can sample short-lived activity; reanalysis and satellite precipitation describe regional environmental context; and rapid maps capture selected downstream consequences. Reviews of High Mountain Asia similarly identify spatial resolution, cloud, terrain, acquisition timing and incomplete end-to-end process observations as central limitations (Kirschbaum et al., 2019).

Our open-data reconstruction recovers first-order source location and changed-surface uncertainty, route geometry, matched hydroclimatic context and operationally mapped impacts from data that can be independently re-accessed or traced. It also identifies hard limits: open optical snapshots did not resolve the hours-scale activity described externally; radar amplitude did not yield a deformation mechanism; regional climate did not resolve source hydrology; and rapid mapping did not produce a complete field inventory. The absence of unambiguous large-scale change in the prespecified optical sequence is therefore compatible with higher-frequency seismic activity, persistent motion derived from a denser image archive, or changes below the spatial or temporal resolution of the observations.

5.7 Limitations

The principal limitations are: (1) cloud and haze in all early source-covering post-event optical observations; (2) mapped-source envelopes informed by convergent multisensor evidence but not a second independent delineation; (3) no appropriate pre/post DEM and therefore no source, erosion or deposition volume; (4) a representative centreline and smoothed 30 m DEM profile rather than observed particle trajectories; (5) non-specific Sentinel-1 amplitude change and no validated source displacement; (6) a 45.85 h final optical gap; (7) gridded hydroclimate at approximately 0.1° and an elevation mismatch; (8) inability to separate water width and disturbed width; (9) operational rapid maps without field validation or complete-census denominators; and (10) unresolved temporary blockage, material fractions and initiation mechanics.

A contemporaneous study also uses GF imagery, field photography, KKN seismic records, surveillance video, eyewitness timing and a dedicated Sentinel-2 velocity inversion that were not included in our analysis (Guo et al., 2026). Our conclusions therefore concern the constraints available from the analysed open and publicly released record. This narrower evidential scope limits direct statements about entrainment, source composition and short-lived pre-failure activity, while retaining independently reproducible measurements of source change, first-order route geometry, regional climate ranks and mapped impacts.

6 Conclusions

An independent reconstruction from open and publicly released Earth-observation records recovers the first-order source, cascade geometry, environmental context and downstream consequences of the 26 August 2026 Gyirong event. The preferred

post-event changed/source surface is 1.009 km² within conservative–inclusive mapping envelopes of 0.491–1.841 km². A representative source-to-port route extends 21.843 km in plan and descends approximately 3396 m. These values agree in scale with a contemporaneous event-specific reconstruction, while the source-area estimates remain definition- and sensor-dependent and should not be merged. The open record supports classification as a mixed rock–ice slope failure followed by a channel-coupled mass-flow/flood cascade, but it does not independently quantify progressive entrainment, failure volume, rheology or a temporary blockage. Event-specific field, high-resolution, seismic and velocity evidence published elsewhere provides stronger constraints on some of those processes. Our separation of external evidence from analytical inputs preserves the independence and reproducibility of the present results.

The prespecified pre-event Sentinel-2 sequence showed no unambiguous large-scale surface change, and the final usable optical observation preceded failure by 45.85 h. This finding is limited by 10 m sampling and variable cloud, snow and illumination; it neither demonstrates stability nor excludes persistent motion, sub-pixel or obscured change, or hours-scale activity recoverable with denser or higher-frequency observations. Neither gridded precipitation product indicated an unusually wet antecedent period within the five prespecified windows. Regional thermal conditions were unusually warm: the 7 d ERA5-Land mean temperature and PDD ranked above all matched 2001–2025 years. These climate observations provide environmental context, not causal attribution. Independent rapid mapping delineated a 37.353 km² downstream disturbed footprint and contained 695 building points and 15.671 km of roads graded destroyed, but the layers are preliminary rather than a complete census. Historical GHSL data independently demonstrate substantial growth of mapped built-up exposure in the affected downstream zones, while the source-image record is insufficient for a quantitative multidecadal source-cryosphere trend. The present evidence supports anthropogenic influence on regional climate and substantial growth of downstream built-up exposure, but it does not establish a direct anthropogenic cause of the 2026 Gyirong slope failure.

The transferable conclusion is that open Earth observation can reconstruct the first-order source–pathway–receptor chain of a rapidly evolving high-mountain cascade while making observation-scale limits explicit. Regional anthropogenic climate influence is not event attribution; non-detection of large persistent local disturbance is not proof that no human activity occurred; lack of visible change in the prespecified optical sequence is not evidence of no pre-failure motion; and historical exposure growth is not hazard amplification. Source perimeters, short-lived activity and causal mechanisms remain dependent on sensor definition, cadence and complementary field or geophysical evidence.

Code and data availability. Sentinel-2 and Sentinel-1 data are available from the Copernicus Data Space Ecosystem; Landsat Collection 2 data are available from the U.S. Geological Survey; ERA5-Land is available from the Copernicus Climate Data Store (<https://doi.org/10.24381/cds.e2161bac>); GPM IMERG is available from NASA GES DISC (<https://doi.org/10.5067/GPM/IMERG/3B-HH-L/07>); Berkeley Earth monthly land-temperature data are available from Berkeley Earth; CMIP6/DAMIP data were obtained from the cloud-optimised Pangeo CMIP6 catalogue; and GHSL R2023A built-up-surface data are available from the European Commission Joint Research Centre. Planet public disaster imagery is subject to its CC BY-NC 4.0 terms and should be accessed from the source catalogue rather than redistributed without confirmation. UNOSAT, Copernicus EMSR927 and OpenStreetMap retain their provider licences and attribution requirements.

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Hydroclimatic conditions before failure

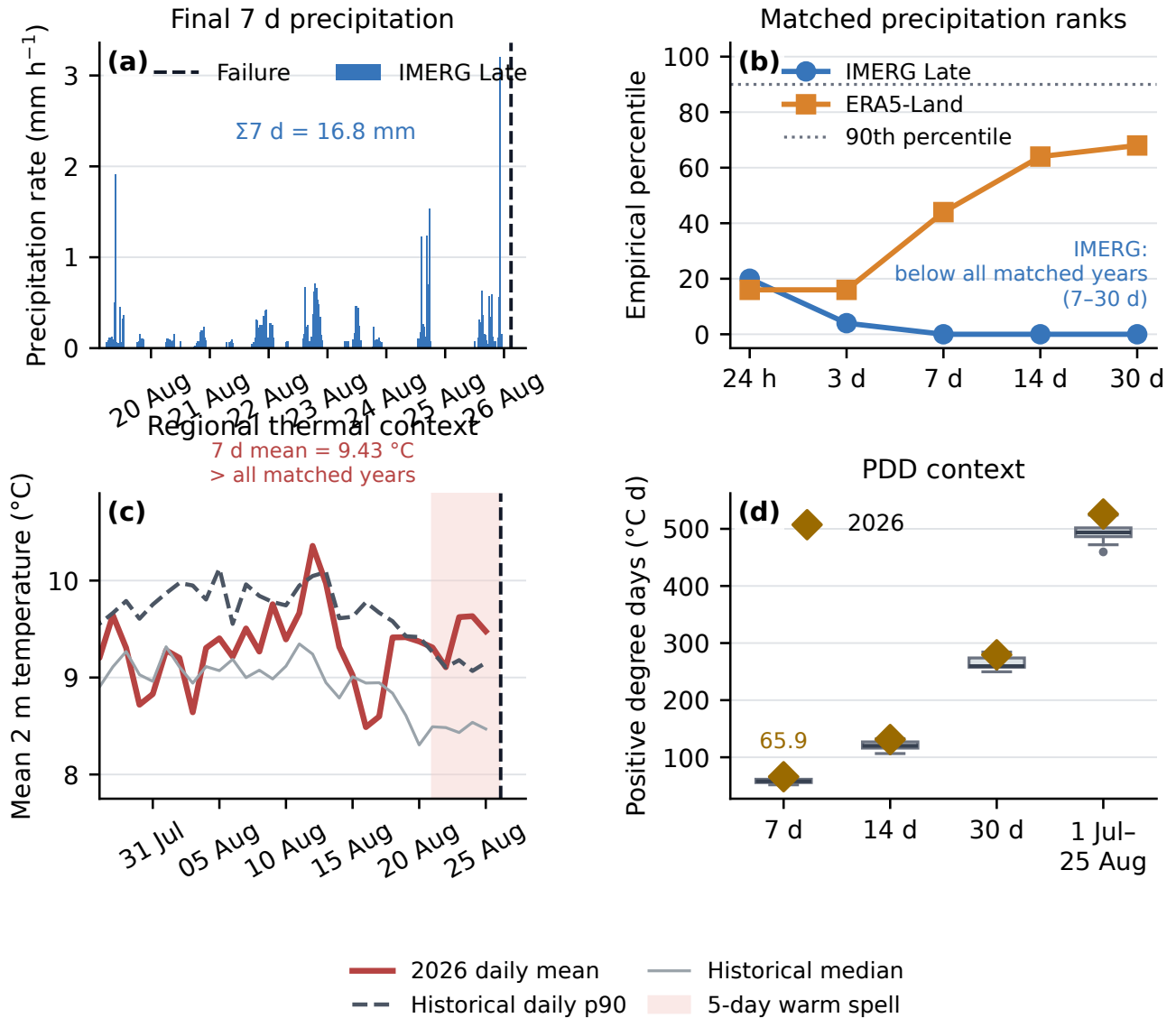


Figure 7. Hydroclimatic conditions preceding failure. (a) GPM IMERG V07 Late half-hourly precipitation during the final 7 d. (b) Empirical precipitation percentiles for five prespecified windows, with the 90th percentile marked; the IMERG 7–30 d ranks were lower than all 25 matched years. (c) ERA5-Land daily mean 2 m temperature, matched historical median and daily 90th-percentile threshold; shading marks the five-day warm spell ending 25 August. (d) Unadjusted PDD totals in 2026 relative to the 25 matched historical years. Dashed lines mark failure at 02:52 UTC. The 7 d temperature and PDD exceeded all matched years, whereas neither precipitation product indicated unusually wet conditions. These are regional environmental data, not trigger attribution.

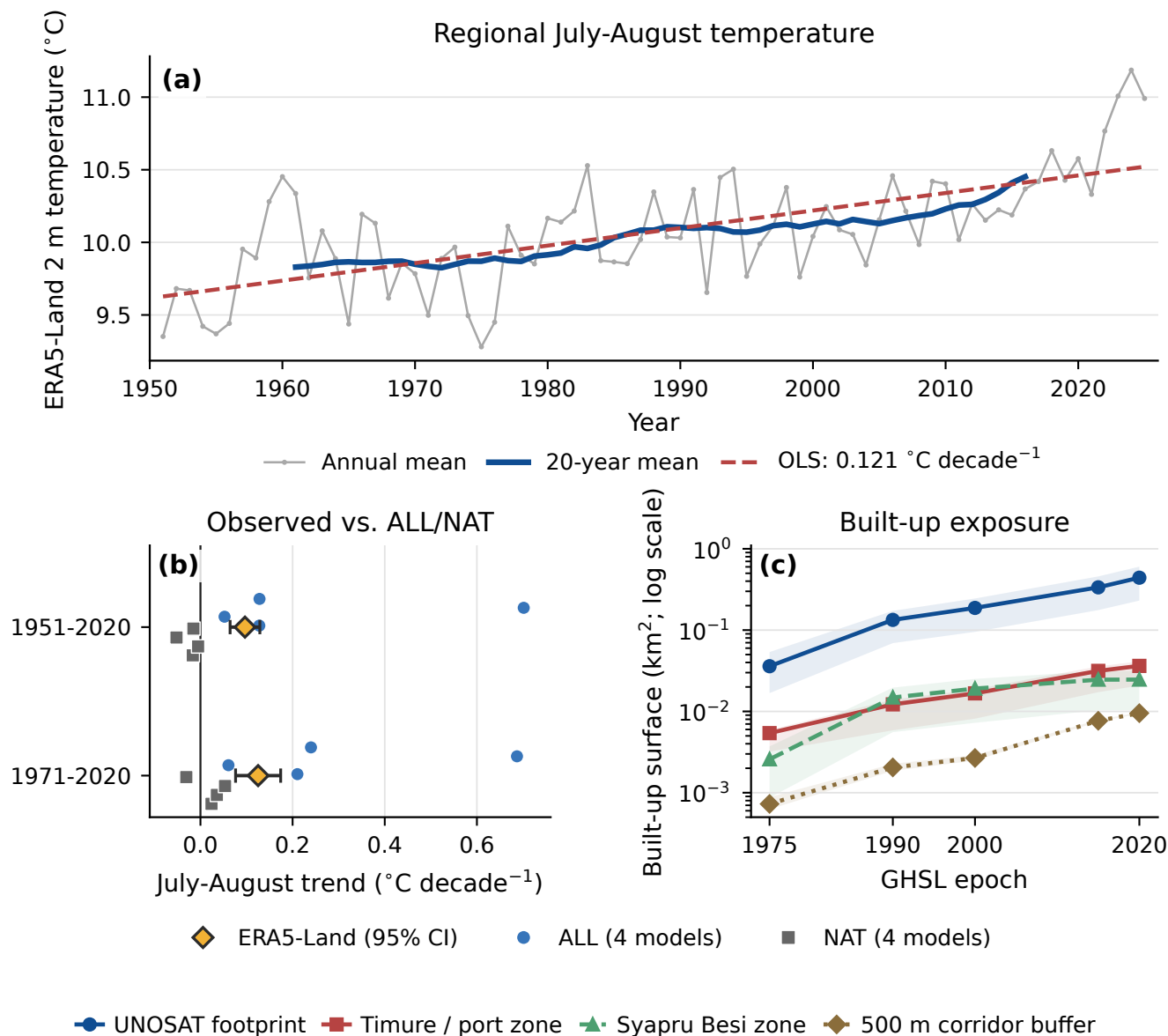


Figure 8. Historical regional temperature, forcing consistency and downstream built-up exposure. (a) Annual July–August ERA5-Land regional mean temperature, 20-year running mean and OLS trend for 1951–2025. (b) Observed ERA5-Land trends with 95 % confidence intervals compared with one-member-per-model July–August CMIP6/DAMIP all-forcing and natural-only trends from four paired models. The comparison indicates forcing consistency and is not formal event attribution. (c) GHSL R2023A mapped built-up surface in four fixed downstream zones for 1975–2020; bands show boundary-sampling sensitivity and the logarithmic axis accommodates the different zone magnitudes. GHSL measures built-up exposure, not population, damage or hazard magnitude. Data: ERA5-Land, CMIP6/DAMIP and GHSL R2023A.

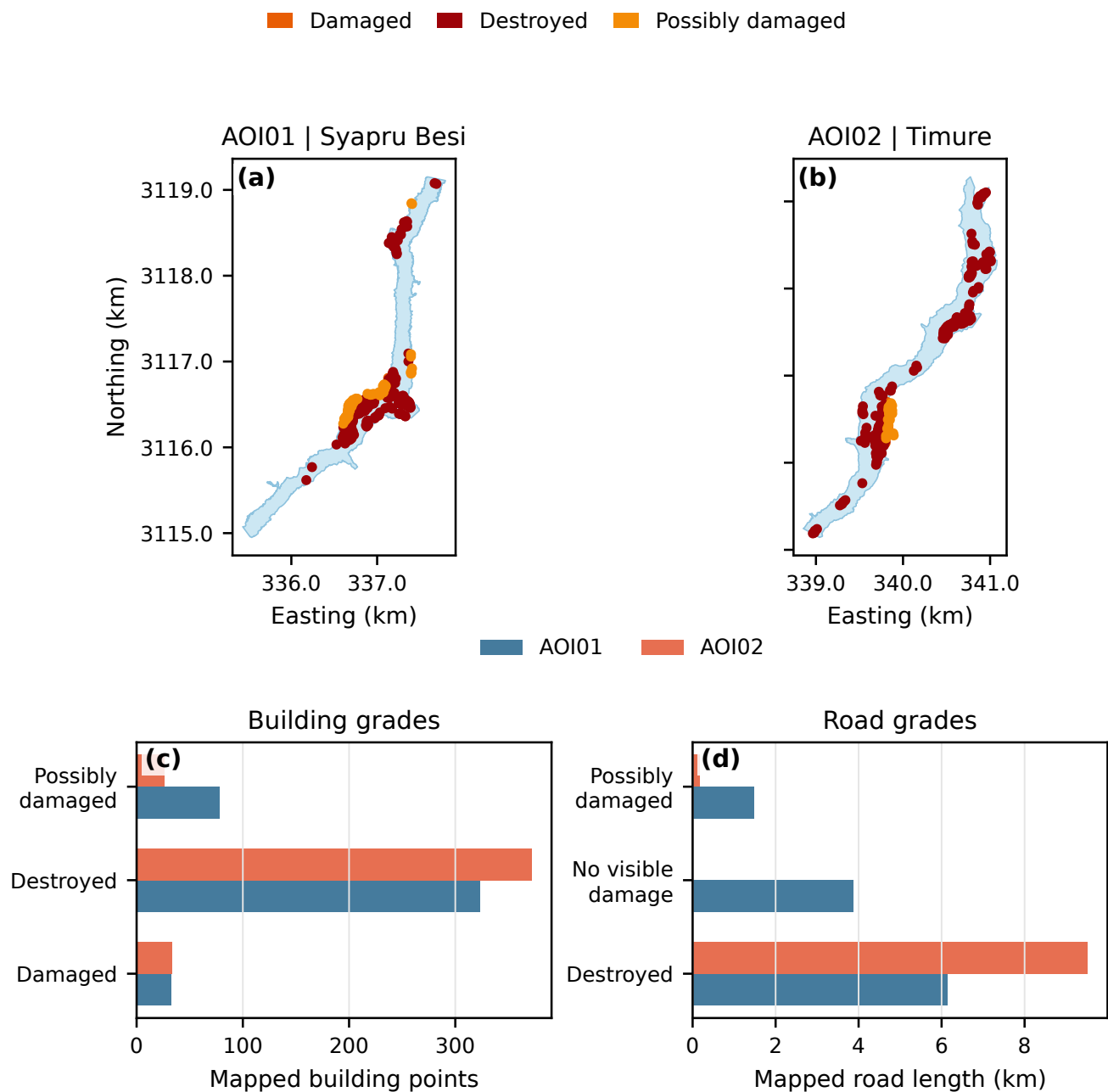


Figure 9. Independent rapid-mapping evidence for downstream impacts. Copernicus EMSR927 maps affected buildings and roads in Syapru Besi and Timure/Gyirong Port. Across the published layers, 695 building points and 15.671 km of roads were graded destroyed. Counts describe preliminary mapped features only and are not a complete census. Data: Copernicus EMSR927 and UNOSAT.